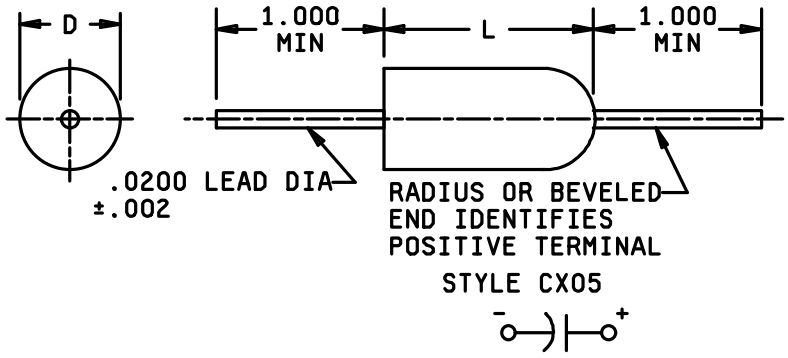
INCH-POUND
MIL-PRF-49137/5A
9 October 1997
SUPERSEDING
MIL-C-49137/5
14 May 1979

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM, POLAR, MOLDED, NONHERMETICALLY SEALED, STYLE CX05

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-PRF-49137.



CIRCUIT DIAGRAM

Case size	Dimensions	
	Diameter D ±.015 (.38)	Length L ±.020 (.51)
A	.085 (2.16)	.250 (6.35)
B	.100 (2.54)	.280 (7.11)
C	.170 (4.32)	.335 (8.51)
D	.170 (4.32)	.410 (10.41)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Metric equivalents are in parentheses.

FIGURE 1. Dimensions and configuration.

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TABLE I. Style CX05 characteristics.

Type designation 1/	DC rated voltage (85°C)	Cap. (nom)	DC leakage (max) +25°C	Dissipation factor	Case size
	Volts	µF	µA	%	
CX05D475-	6	4.7	1.0	6	A
CX05D156-	6	15	1.0	8	B
CX05D336-	6	33	1.5	8	C
CX05D476-	6	47	3.0	8	D
CX05F335-	10	3.3	1.0	6	A
CX05F106-	10	10	1.0	8	B
CX05F226-	10	22	1.5	8	C
CX05F276-	10	27	2.0	8	D
CX05F336-	10	33	3.0	8	D
CX05F396-	10	39	5.0	8	D
CX05F476-	10	47	5.0	8	D
CX05H225-	15	2.2	1.0	6	A
CX05H685-	15	6.8	1.0	6	B
CX05H156-	15	15	1.5	8	C
CX05H226-	15	22	3.0	8	D
CX05H336-	15	33	5.0	8	D
CX05J155-	20	1.5	1.0	6	A
CX05J475-	20	4.7	1.0	6	B
CX05J126-	20	12	1.0	8	D
CX05J156-	20	15	3.0	8	D
CX05K105-	25	1.0	1.0	4	A
CX05K225-	25	2.2	1.0	6	B
CX05K335-	25	3.3	1.0	6	B
CX05K685-	25	6.8	1.5	6	C
CX05K106-	25	10	1.5	8	C
CX05M334-	35	0.33	1.0	4	A
CX05M474-	35	0.47	1.0	4	A
CX05M155-	35	1.5	1.0	6	B
CX05M335-	35	3.3	1.5	6	C
CX05M395-	35	3.9	1.5	6	C
CX05M475-	35	4.7	1.5	6	C
CX05M685-	35	6.8	3.0	6	D
CX05M106-	35	10	5.0	8	D
CX05N104-	50	0.10	1.0	4	A
CX05N154-	50	0.15	1.0	4	A
CX05N224-	50	0.22	1.0	4	A
CX05N334-	50	0.33	1.0	4	B
CX05N474-	50	0.47	1.0	4	B
CX05N684-	50	0.68	1.0	4	B
CX05N105-	50	1.0	1.0	4	B
CX05N155-	50	1.5	1.5	6	C
CX05N225-	50	2.2	1.5	6	C
CX05N335-	50	3.3	2.0	6	D
CX05N475-	50	4.7	3.0	6	D

1/ Complete type designation will include an additional symbol indicating capacitance tolerance.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.
Case material: Molded, epoxy or plastic.
Leads: Solder-coated metal.
DC rated voltage: See table I.
Operating temperature: -55° to +85°C.

DC leakage (DCL): See table I.

Capacitance (cap.): See table I.
Cap. tolerance: ±10% (K), or ±20% (M).

Dissipation factor (DF): See table I.

Stability at low and high temperature: In accordance with MIL-PRF-49137.

Step 1 (+25°):	DCL	- See table I.
	Cap.	- Within tolerance specified in table I.
	DF	- See table I.
Step 2 (-55°C):	ΔCap.	- Within -15 percent of step 1 measured value.
	DF	- 200 percent of table I.
Step 3 (+25°C):	DCL	- See table I.
	ΔCap.	- Within 10 percent of step 1 measured value.
	DF	- See table I.
Step 4 (+85°C):	DCL	- 10 times 25°C limit (see table I).
	ΔCap.	- Within +15 percent of step 1 measured value.
	DF	- 150 percent of table I.
Step 5 (+25°C):	DCL	- See table I.
	ΔCap.	- Within 10 percent of step 1 measured value.
	DF	- See table I.

Surge voltage: In accordance with MIL-PRF-49137.
DCL - See table I.
ΔCap. - Within 10 percent of initial value.
DF - See table I.

Life: Method 108 of MIL-STD-202.
1,000 hours:

At 25°C:	DCL	- 200 percent of table I.
	ΔCap.	- Within 10 percent of initial measured value.
	DF	- See table I.

NOTES:
1. Revision letters are not used to denote change due to the extensiveness of the changes.
2. These capacitors are intended to be used only where supplemental moisture protection is provided or for noncritical applications where hermetic moisture protection is not required.

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Custodians:
Army - CR
Navy - EC
Air Force - 11

Review activities:
Navy - AS, CG, MC, OS SH
Air Force - 19, 85, 99
DLA - ES

Preparing activity:
Army - CR

Agent:
DLA - CC

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